Marius Arcay

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Education

San Jose State University — **San Jose, CA**
Bachelor of Science in Computer Science — Spring 2026 – Spring. 2028 (Expected)

West Valley College — **Saratoga, CA**
Associate of Science in Physics — Winter 2026 – Spring 2028 (Expected)
Associate of Science in Mathematics — Fall 2022 – Winter 2026

Experience

Contract 3D Artist — Oct 2025 – Jan 2026
Valve Corporation — *Remote*

- Authored **Rooftop**, an official Counter-Strike 2 multiplayer map using Valve's proprietary technology, balancing visual fidelity, gameplay requirements, and performance
- Profiled and optimized scene geometry, materials, lighting, and assets to meet performance standards expected of an official Counter-Strike 2 map
- Shipped to a global audience of millions of players

Retail Tech Consultant — Jun 2021 – Present
Target — *Morgan Hill, CA*

- Provided personalized expertise to help guests select, understand, and purchase electronics

Projects

CUDA Fractal Renderer | *C++, CUDA, Multithreading, Git, Visual Studio* — June 2026 – July 2026

- Built a CUDA fractal renderer achieving a 100× GPU speedup over single-threaded CPU benchmarks across Mandelbrot and Julia sets
- Developed a benchmarking system comparing single-threaded CPU, multi-threaded CPU, and CUDA rendering speeds, alongside dynamic iteration scaling, click-and-drag interactions, and a video/animation rendering pipeline
- Automated Visual Studio project configuration to detect unsupported CUDA environments and remove CUDA-specific dependencies during compilation

OpenGL PBR Path Tracer | *C++, OpenGL, GLSL, Git, Visual Studio* — May 2026 – June 2026

- Built a GPU-accelerated path tracer using OpenGL/GLSL with Monte Carlo sampling and Cook-Torrance GGX PBR, supporting albedo, normal, roughness, and metallic maps
- Implemented a depth prepass to reduce overdraw alongside scene and model importing, texture pooling, bounding-volume intersection testing, animation system, and image/video rendering
- Developed interactive rendering tools including a translation gizmo, real-time material editing, debug visualization modes, and scene manipulation controls

3D Graphics Library | *C++, Multi-threading, Visual Studio* — Nov 2025 – March 2026

- Built a multi-threaded CPU 3D renderer in C++
- Implemented a model conversion pipeline, simple animation system, and flexible scene configuration

Environmental Monitoring Device | *C, MySQL, I2C* — April 2024 – May 2024

- Developed a Raspberry Pi-based device in C to collect temperature, humidity, and luminance measurements through I2C sensors
- Implemented MySQL-based logging and configurable data visualization for long-term environmental monitoring

Technical Skills

Languages: C++, C, GLSL, Python, Java, Assembly (ARM)
Graphics & GPU: OpenGL, CUDA, PBR, Monte Carlo Rendering
Systems: Multithreading, Linux, Windows, I2C
Libraries & Tools: OpenCV, raylib, Git, Visual Studio, VS Code